

Revised Generation Interconnection System Impact Study Report

For Queue Project Q20-1545

Chatom 138 kV
October 17, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1545. The Interconnection Customer proposes a 636.1 MW solar photovoltaic facility interconnecting at the Chatom 138 kV substation in the Virginia area. The request was received on August 14, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 542 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$73,090,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Mitchell Gardens - Kapstone 138 KV Line	Purdy - Page 138 KV	164.5	501	20.3
Templeton - Buckhorn 138 KV	Roslyn - S Bluefield 138 KV L	146.3	172	9.7
Frito Lay - Timberville 138 KV	Purdy Substation - Chatom 138 KV L	146.1	322	23.9
New Salisbury - Timberville 138 KV	Purdy - Chatom 138 KV Line	145.1	328	23.9
Roslyn - Buckhorn 138 KV	S Bluefield - Roslyn 138 KV	143.5	177	9.7
Cherry Substation - Mount Jackson 138 KV	Dpc Prairie Du Chien - Frito Lay Substation 138 KV Line	143.2	254	12.3
Lorin Industries - Tampa Bay 138 KV line	Dpc Prairie Du Chien - Tampa Bay 138 KV	135.2	292	13.8
Templeton - Red Tail 138 KV	S Bluefield - Roslyn 138 KV Line	134.5	186	9.7
Dpc Prairie Du Chien - Cherry 138 KV L	Dpc Prairie Du Chien - Frito Lay 138 KV	134.2	264	12.3
Alvin - Nuclear Fuels 138 KV	Purdy - Page 138 KV	133.3	310	30.1
Dpc Prairie Du Chien - Lorin Industries 138 KV line	Dpc Prairie Du Chien - Tampa Bay 138 KV	133.1	305	13.8
Tes Filer City Sta - Einburg 138 KV	Page - Purdy 138 KV	132.6	262	12.5
Tampa Bay - Roslyn 138 KV	Purdy - Page 138 KV line	131.7	427	19.8
Edinburg - Springfield Power Llc 138 KV	Purdy - Page 138 KV	130.4	275	12.5
Tampa Bay - Dpc Prairie Du Chien 138 KV	Purdy - Page 138 KV Line	130.4	353	15.9

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
St. Martins Substation - Wadsworth 138 KV	Nuclear Fuels - St Louis 138 KV	130.1	210	20.3
S Bluefield - Springfield Power Llc 138 KV	Purdy - Page Substation 138 KV	127.2	297	12.5
Roslyn - S Bluefield 138 KV	Purdy Substation - Page 138 KV	126.1	305	12.5
Dpc Prairie Du Chien - Frito Lay 138 KV	Purdy - Catom 138 KV	125.9	476	22.1
Chatom - Alvin 138 KV line	Purdy - Page 138 KV	124.0	278	30.1
St Barnabas - Folsom 138 KV	Bath County - Sherburne 345 KV	120.5	255	8.0
Plant City - Red Tail 138 KV	Roslyn Substation - South Bluefield 138 KV	119.1	203	9.7
Detour - Cedarville Junction 138 KV Line	Mitchell Gardens - Kapstone 138 KV	119.0	566	22.0
Cedarville Junction - Page 138 KV	Kapstone - Mitchell Gardens 138 KV	118.8	571	22.0
Detour - Siloam Springs 138 KV	Detour - Fort Benton 138 KV line	118.6	460	14.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Mitchell Gardens - Kapstone 138 KV L	Rebuild line with higher-capacity conductor	\$3,434,000	40% (\$1,368,000)
Templeton - Buckhorn 138 KV line	Rebuild line with higher-capacity conductor	\$4,317,000	46% (\$1,990,000)
Timberville - Frito Lay 138 KV	Rebuild line with higher-capacity conductor	\$4,327,000	47% (\$2,052,000)
New Salisbury - Timberville 138 KV	Rebuild line with higher-capacity conductor	\$7,508,000	47% (\$3,561,000)
Roslyn - Buckhorn 138 KV	Rebuild line with higher-capacity conductor	\$5,543,000	46% (\$2,555,000)
Cherry Substation - Mount Jackson 138 KV	Rebuild line with higher-capacity conductor	\$6,930,000	38% (\$2,667,000)
Tampa Bay - Lorin Industries 138 KV	Rebuild line with higher-capacity conductor	\$6,408,000	43% (\$2,741,000)
Templeton - Red Tail 138 KV	Rebuild line with higher-capacity conductor	\$13,044,000	46% (\$6,013,000)
Dpc Prairie Du Chien - Cherry 138 KV L	Rebuild line with higher-capacity conductor	\$9,622,000	38% (\$3,703,000)
Nuclear Fuels - Alvin 138 KV line	Rebuild line with higher-capacity conductor	\$5,720,000	53% (\$3,046,000)
Dpc Prairie Du Chien - Lorin Industries 138 KV line	Rebuild line with higher-capacity conductor	\$2,066,000	43% (\$884,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Tes Filer City Sta - Einburg 138 KV	Rebuild line with higher-capacity conductor	\$11,184,000	47% (\$5,206,000)
Tampa Bay - Roslyn 138 KV	Rebuild line with higher-capacity conductor	\$7,825,000	43% (\$3,338,000)
Edinburg - Springfield Power Llc 138 KV	Rebuild line with higher-capacity conductor	\$4,340,000	47% (\$2,021,000)
Tampa Bay - Dpc Prairie Du Chien 138 KV	Rebuild line with higher-capacity conductor	\$3,141,000	46% (\$1,439,000)
St. Martins Substation - Wadsworth 138 KV	Rebuild line with higher-capacity conductor	\$4,820,000	46% (\$2,191,000)
South Bluefield - Springfield Power Llc 138 KV	Rebuild line with higher-capacity conductor	\$3,043,000	47% (\$1,416,000)
Roslyn - S Bluefield 138 KV	Rebuild line with higher-capacity conductor	\$7,696,000	47% (\$3,583,000)
Dpc Prairie Du Chien - Frito Lay 138 KV	Rebuild line with higher-capacity conductor	\$4,519,000	31% (\$1,412,000)
Chatom - Alvin 138 KV line	Rebuild line with higher-capacity conductor	\$4,688,000	51% (\$2,383,000)
St Barnabas - Folsom 138 KV	Rebuild line with higher-capacity conductor	\$9,801,000	17% (\$1,642,000)
Plant City - Red Tail 138 KV	Rebuild line with higher-capacity conductor	\$8,542,000	46% (\$3,938,000)
Detour - Cedarville Junction 138 KV L	Rebuild line with higher-capacity conductor	\$9,680,000	40% (\$3,880,000)
Page - Cedarville Junction 138 KV L	Rebuild line with higher-capacity conductor	\$5,983,000	40% (\$2,398,000)
Siloam Springs - Detour 138 KV	Rebuild line with higher-capacity conductor	\$23,937,000	32% (\$7,663,000)
	Total allocated to project		\$73,090,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.